

Experiment (7)

MOMENT OF INERTIA

7.1 Introduction

Moment of Inertia is a quantity expressing a rigid body's tendency to resist angular acceleration. You can think of it as an angular mass property that links torque to angular acceleration in the same manner that “regular” mass links force to linear acceleration. Unlike mass, of course, it cannot be measured simply by using a balance. Since it describes resistance to rotational acceleration, it must be defined relative to a rotation axis. In this experiment, we will learn how we can take a quantitative measurement of the Moment of Inertia. We will use the measurements to test the rotational analogue of Newton’s Second Law of Motion.

7.2 Objectives

- Investigate and measure quantities that describe the Planar Rotational Motion of a Rigid Body (angular position, angular velocity and angular acceleration)
- Verify the rotational analogue of Newton’s second law of motion (relation between torque, moment of inertia and angular acceleration).
- Use Cobra4 data acquisition system, the light barrier, and the data analysis software to collect and analyze data.

7.3 Theory

A rigid body is the ideal case of a solid body in which deformation is neglected regardless of the external forces exerted on it.

Consider a rigid disk starting to rotate anticlockwise about a fixed axis normal to the plane of the disk and passing through the center o as shown in Figure 7.1. A line passing from the point o to the point P_0 at time $t = 0$ will progress to the new position oP_1 during a time t , making an angle θ (between the line oP_1 and the x-axis).

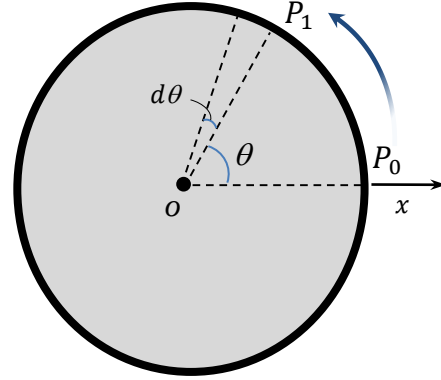


Figure 7.1: A rotating rigid disk.

The angle θ (in radian) is called the *angular displacement* of the point at time t . If the disk at this instance rotates a small angle $\Delta\theta$ during a time Δt , the angular velocity ω is given by,

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt} \quad (7.1)$$

and the angular acceleration is defined by,

$$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} \quad (7.2)$$

Note that the angular velocity ω is assumed to be positive when the cylinder rotates anticlockwise, according to the “right-hand rule”.

For a point at P_1 rotating in a circular path of radius r , the relation between the tangential speed v and the angular speed ω is given by,

$$v = \omega r \quad (7.3)$$

and the relation between the tangential acceleration a and the angular acceleration α is given by,

$$a = \alpha r \quad (7.4)$$

To rotate the disk about its axis, a torque τ (caused by a force in the plane of the cylinder that does not pass through the axis of rotation) must be applied. According to Newton's second law of rotational motion, the relation between the applied torque τ and the angular acceleration of the rotating body is given by,

$$\tau = I\alpha \quad (7.5)$$

where I is called the moment of inertia of the body about its axis of rotation. It is a quantity expressing the body's tendency to resist angular acceleration. In order to acquire the same angular acceleration, a body having a large moment of inertia about a certain axis will need a larger torque than that required for another body with smaller moment of inertia about the same axis of rotation. The [Appendix](#) shows the moment of inertia of some geometrical shapes.

In this experiment, the moment of inertia of some simple bodies are measured. As shown in Figure 7.2, a mass M is used to rotate the rigid body (the disk for example) about its axis using a silk thread, which is wound about a pulley of radius r_p .

If the tension in the thread is T , the torque τ acting on the pulley and the disk is given by,

$$\tau = T r_p \quad (7.6)$$

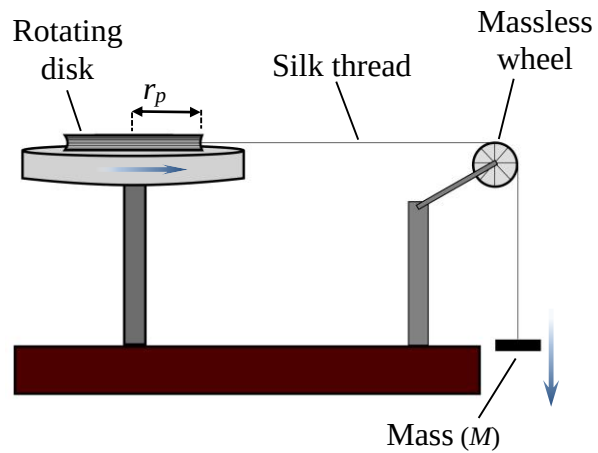


Figure 7.2: Simple diagram for the experiment setup

Applying Newton's second law on the moving mass M we get

$$Mg - T = Ma \quad (7.7)$$

where a is the acceleration of the moving mass and also the tangential acceleration of the pulley's rim on which the thread is wound. Using equations (7.4) to (7.7) we get,

$$I = \frac{\tau}{\alpha} = \frac{M(g - \alpha r_p)r_p}{\alpha} \quad (7.8)$$

If the angular acceleration α is measured experimentally, we can calculate the moment of inertia of the body about the axis of rotation using relation (7.8).



In this experiment, the Cobra4 data acquisition system and the wheel incremental encoder are used to collect data and measure the angular displacement of the rotating body as a function of time $\theta_p(t)$. The wheel incremental encoder can only measure the angular displacement of the wheel itself as a function of time $\theta_w(t)$. Figure 7.3 shows the hanging mass as it moves downward a certain distance and the thread unwinds from the pulley and moves over the wheel without slipping.

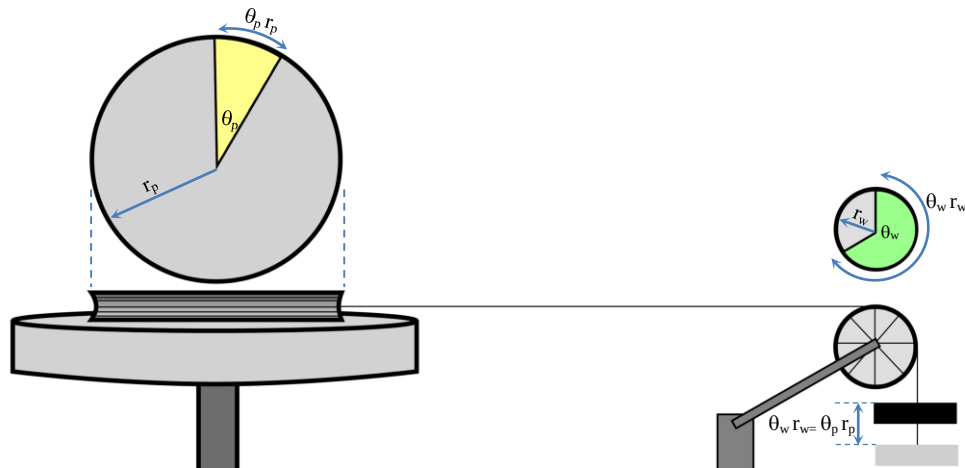


Figure 7.3: The thread unwinds from the pulley and moves over the wheel without slipping.

In this case, the covered perimeter of the pulley $\theta_p r_p$ and that covered of the wheel $\theta_w r_w$ are equal. The radius of the wheel is already known to the Measure software and the user inputs the radius of the pulley. The Measure software then uses these two radii to compute the angular displacement of the rotating body.

$$\theta_{rot. \atop body}(t) = \theta_p(t) = \frac{r_w}{r_p} \theta_w(t) \quad (7.9)$$

The angular acceleration of the rotating body can then be calculated by differentiating the angular displacement twice with respect to time.

$$\alpha = \frac{d^2 \theta_p(t)}{dt^2} \quad (7.10)$$

7.4 Setup

The moment of inertia setup main parts are shown in Figure 7.4.

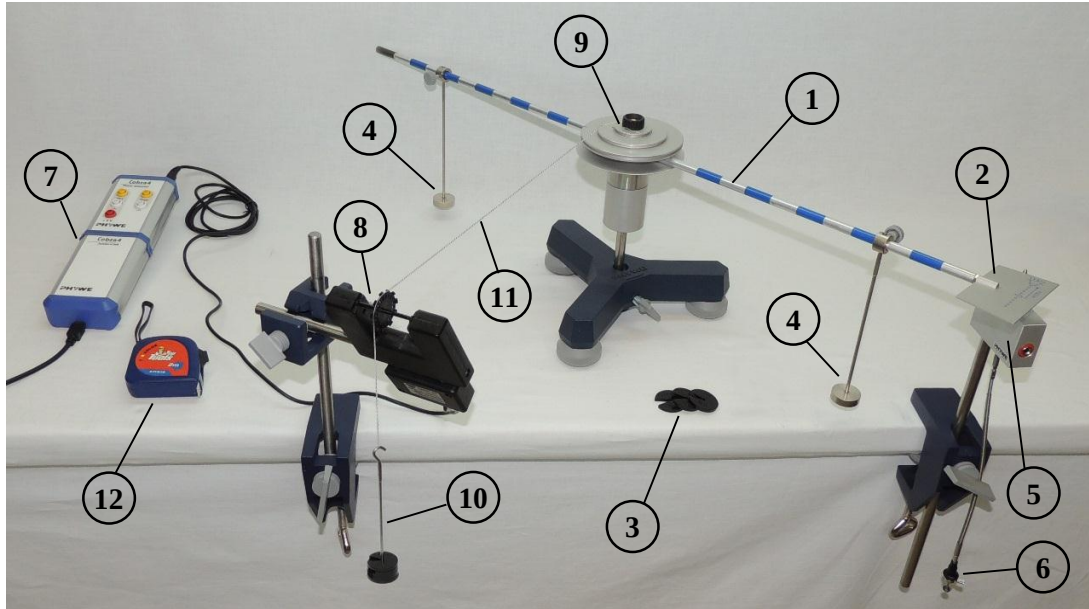


Figure 7.4: Moment of inertia setup main parts.

- | | |
|-------------------------------------|--|
| 1 Inertia rod | 7 Cobra4 Sensor-Unit Timer/Counter |
| 2 Inertia rod holding plate | 8 Wheel incremental encoder |
| 3 Slotted masses | 9 Axle with three pulleys of different radii |
| 4 Inertia rod weight holders, 50 gm | 10 Weight holder |
| 5 Holding device | 11 Silk thread |
| 6 Release trigger | 12 Measuring tape |

7.5 Procedure

7.5.1 Task (1): Determination of the moment of inertia of a bar with constant mass mounted to it at different distances from the axis of rotation.

- 1- Set up the experiment and set the connections as shown in Figure 7.4 above.
- 2- On the computer desktop, click the “Measure” icon to start the software. If you have connected the USB to the computer properly, the startup screen of the Measure software will appear as shown in Figure 7.5.

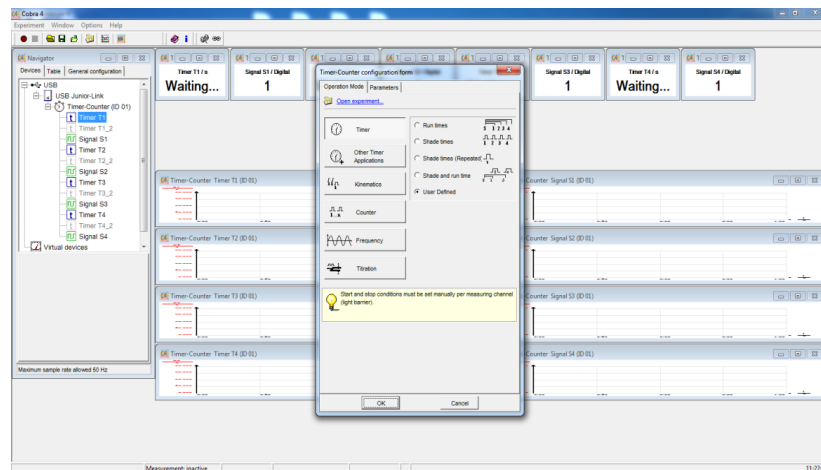


Figure 7.5: Measure Software startup screen.

- 3- Click “Cancel” then go to the “Experiment” menu and select “Load configuration”. From the window shown in Figure 7.6, You will find the experiment configuration file in the “Angular acceleration” folder found on the desktop. Select the configuration file for the rod then click “OK” to load the experiment configuration.

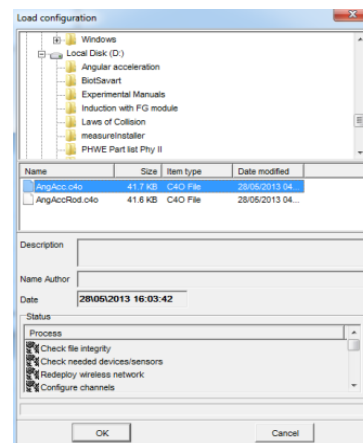


Figure 7.6: Loading the configuration file.



Make sure that one end of the silk thread (11) is attached to the axle (9) of rotation of the inertia rod (1). If not, ask a TA for help.

- 4- Lay the silk thread (11) across the wheel (8) of the incremental encoder as shown in Figure 7.7 then add 90 gm slotted masses (3) to the weight holder (10) to have a total of 100 gm hanging freely from the silk thread.
- 5- Unscrew and remove both the inertia rod weight holders (4) as shown in Figure 7.8 to calculate the moment of inertia of the rod without any other added extra masses. (You will need to unmount the inertia rod holding plate (2) in order to remove the inertia rod weight holders)
- 6- Wind the thread several times around one of the axle (9) pulleys shown in Figure 7.9 enough turns until the upper tip of the weight holder is about **5 cm** below the wheel as shown in Figure 7.7 above. One should know the radius of the used pulley r_p as it will be used in the measure software.



Figure 7.7: Laying the thread across the wheel.

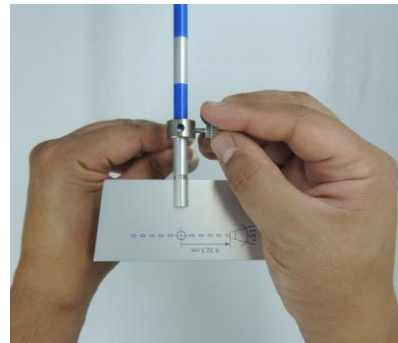


Figure 7.8: Unscrewing an inertia rod weight holder.

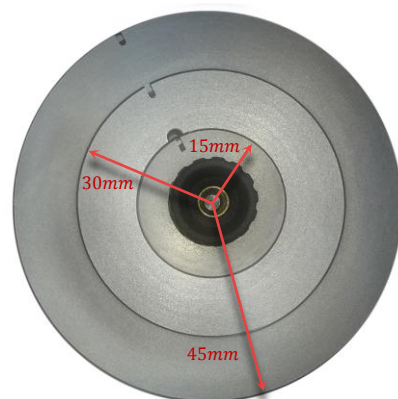


Figure 7.9: Axle pulleys.



Make sure that the silk thread is long enough such that the weight holder reaches the ground after it is totally unwound. If not, ask your TA to replace it for you.



Make sure that the silk thread is making a 90° angle with the axis of the wheel as shown in Figure 7.10. Otherwise, the thread may fall off the wheel during the measurement.

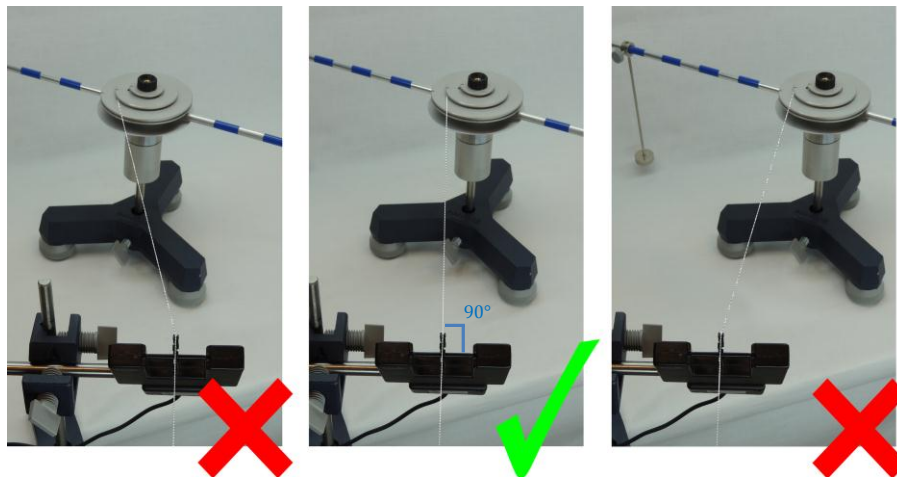


Figure 7.10: Correct and incorrect angles between the silk thread and the axis of the wheel.

Also, make sure that the thread between the axle pulley and the encoder wheel is horizontal. You may need to rotate the encoder as shown in Figure 7.11 to achieve this.

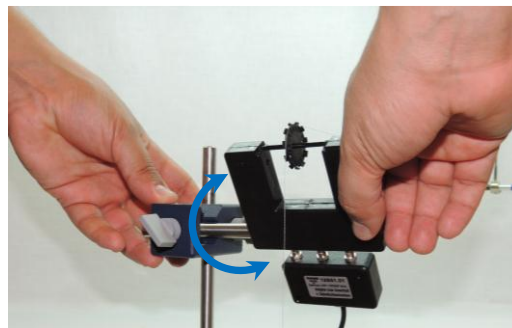


Figure 7.11: Rotating the wheel incremental encoder.

7- Press the release trigger (6) of the the holding device (5) and lock it as shown in Figure 7.12. You will then be able to use

the pin of the holding device and the inertia rod holding plate (2) as shown in Figure 7.13 to prevent the inertia rod from rotating due to the torque applied on it by the weight holder.



Figure 7.12: Locking the release trigger.



Figure 7.13: Preventing the inertia rod from rotating.



Make sure that the holding device pin will not disturb the rotation of the rod after release.

- 8- Start measurement by pressing the record (●) button.
- 9- A window as the one shown in Figure 7.14 will appear to enter the axle pulley radius r_p , that you are using (the one that the thread was wound around in step 6). Enter the value (in **mm**) and press “OK”. At the same instant unlock the release trigger (6) to allow the rod to rotate under the influence of the torque applied on it by the weight holder.
- 10- Stop data recording just before the weight holder reaches the ground by pressing the stop (■) button (in order not to have irrelevant data at the end of your recorded data).

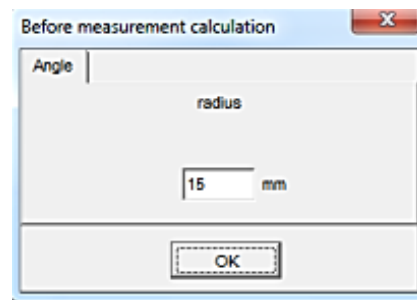


Figure 7.14: Setting the axle pulley radius window.

- 11- Once the recording is terminated, stop the rod from rotating to avoid tangling the thread around the setup.



*Stop the rotating rod by putting your hand on the axle as shown in Figure 7.15. **Never put your hand in the rod's way.***

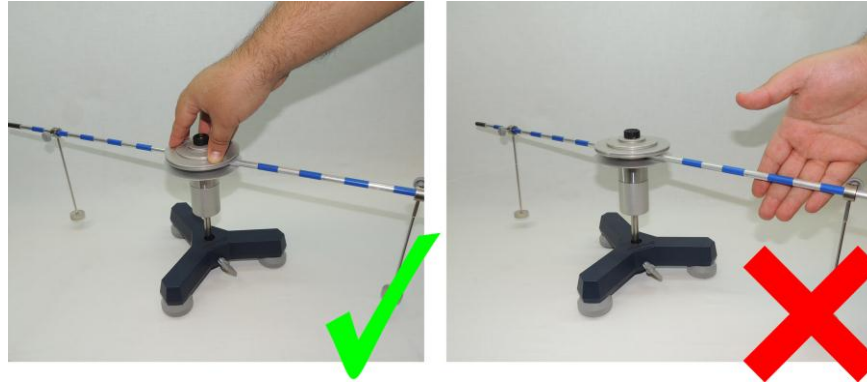


Figure 7.15: Correct and incorrect ways to stop the rotating rod.

- 12- In the measure software, press “send all data to measure” to display and analyze the recorded data as shown in Figure 7.16

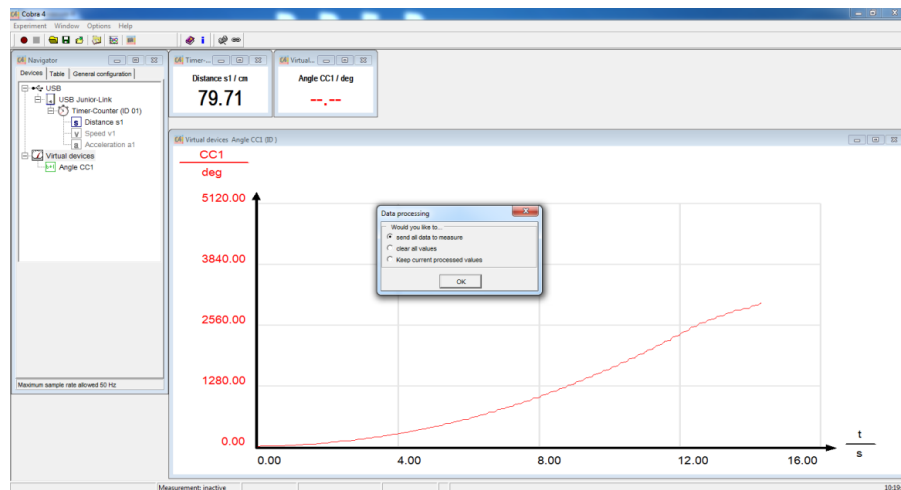


Figure 7.16: Sending the recorded data to be analyzed.

- 13- Use the “Function fitting” tool () to fit the angular displacement curve. Select “parabola”, click “Calculate” then “Add new curve” as shown in Figure 7.17.

Why was the parabola fitting chosen? What do you expect will happen if you choose a polynomial of a different order?

- 14- Click the “Channel modification” tool () , select the fitted curve from the drop down list and select differentiate then click “Calculate” as shown in Figure 7.18 to calculate the angular velocity curve.

- 15- Follow the same procedure of the previous step to calculate the angular acceleration curve by differentiating the angular velocity curve. The three curves of angular displacement, angular velocity and angular acceleration will be as shown in Figure 7.19.

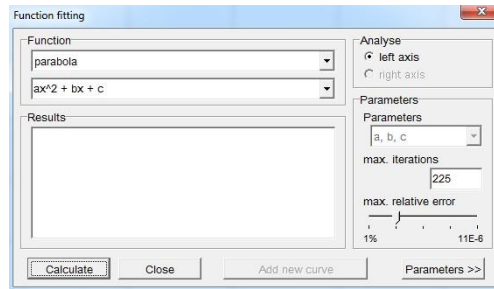


Figure 7.17: Using the function fitting tool.

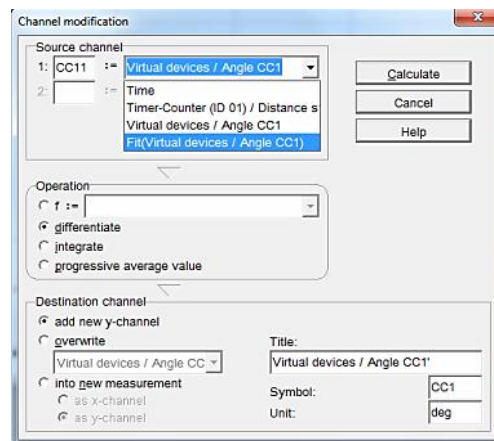


Figure 7.18: Differentiating the angular displacement curve.

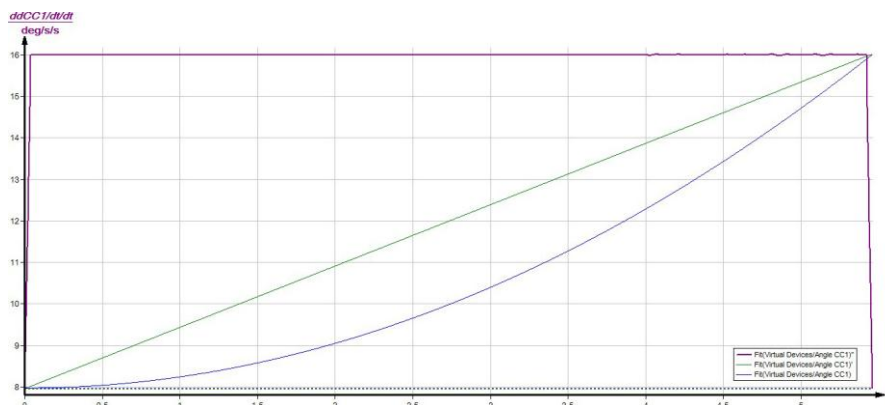


Figure 7.19: Angular displacement, angular velocity and angular acceleration curves.

- 16- Choose the angular acceleration curve by selecting the angular acceleration curve only (**CC1** **CC1** **dCC1/dt** **ddCC1/dt/dt**). Use the “Show average value” tool () which will show a window with the average value of the curve as shown in Figure 7.20.

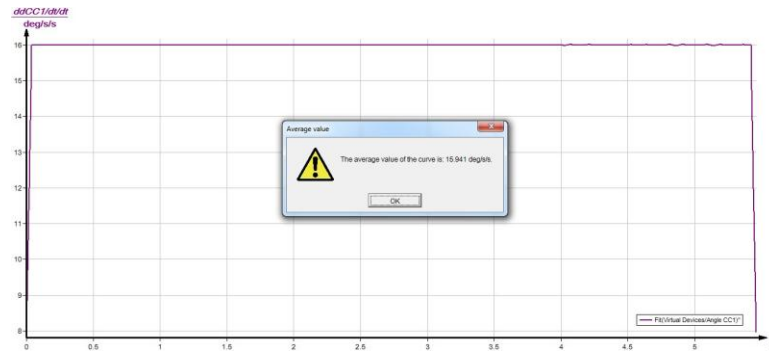


Figure 7.20: The average value of the angular acceleration.



In the average value window, the unit of the average value should be deg/s/s. If not, then the selected curve may not be the angular acceleration curve. Don't forget to convert the value into rad/s/s before any calculations.

- 17- Repeat steps 7 to 16 several times for error analysis then use this data to calculate the moment of inertia of the empty rod using relation (7.8).
- 18- Connect the inertia rod weight holders (4) to the inertia rod (1) again and adjust the distance between the axis of rotation of the rod and each weight holder to be $r = 15 \text{ cm}$ by using the measuring tape (12) as shown in Figure 7.21.

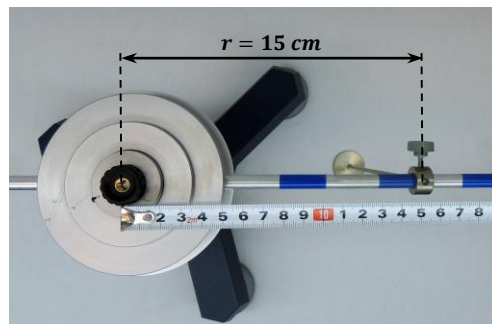


Figure 7.21: Adjusting the position of the inertia rod weight holders.

- 19- Add **50 gm** extra slotted masses (3) to each of the inertia rod weight holders (4) to have a total of **100 gm** hanging from each side of the rod.
- 20- Repeat steps 7 to 16 several times for error analysis.
- 21- Repeat steps 7 to 16 several times for error analysis for each of the following distances between the axis of rotation of the rod and each weight holder **$r = 20 \text{ cm}, 25 \text{ cm}$ and 30 cm** while keeping the total of **100 gm** hanging from each side of the rod.
- 22- Plot a graph (**I vs. r^2**) between the moment of inertia of the rotating system and squared the distance between the axis of rotation of the rod and each weight holders from the axis of rotation.



We can take the moment of inertia of the empty rotating rod (when the inertia rod weight holders were removed) to be the moment of inertia of the rod when the distance between the axis of rotation of the rod and each weight holder $r = 0$.



There is a frictional torque in the bearing of the rotating system that may cause error in your measurements so you need to subtract this error from your measurements to enhance your results.

This frictional torque was measured using very small weights hanging from the silk thread while setting the rod in a free moving position. The small hanging weights are added one by one until the rod starts to rotate, the torque due to the hanging weights (using relation (7.6)) that caused the rod to start rotating equals the frictional torque of the bearing. The frictional torque is found to be approximately $2.94 \times 10^{-4} \text{ N.m}$

Note that this value differs from one setup to another but we may still use it as an approximated value.

7.5.2 Task (2): Determination of the moment of inertia of a bar with varying masses mounted to it at the same distance from the axis of rotation.

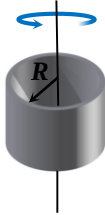
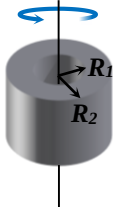
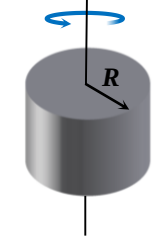
- 1- Return the inertia rod weight holders (4) to be at distance $r = 20 \text{ cm}$ from the axis of rotation of the inertia rod (1).
- 2- Repeat steps 7 to 16 from Task (1) several times for error analysis for each of the following extra slotted masses (3) added to each of the inertia rod weight holders (4) $m_{\text{extra}} = 10 \text{ gm}, 20 \text{ gm}, 30 \text{ gm}, 40 \text{ gm}$ and 50 gm .
- 3- Plot a graph (I vs. m) between the moment of inertia of the rotating system and the total mass hanging from each side of the rod (mass of the weight holder and the extra added masses $m = m_{\text{weight holder}} + m_{\text{extra}}$).

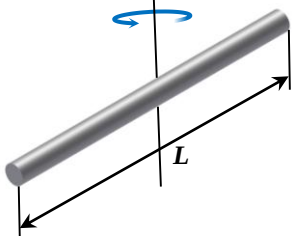
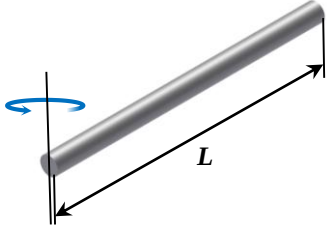
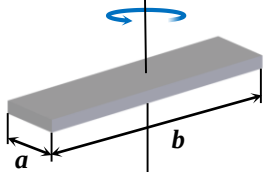


We can take the moment of inertia of the empty rotating rod (when the inertia rod weight holders were removed) calculated in Task (1) to be the moment of inertia of the rod when the total hanging mass $m = 0$.

Appendix

Moment of inertia of some geometrical shapes

Geometrical shape	Moment of inertia about the defined axis
 Cylindrical shell of radius R and mass m rotates about its axis	$m R^2$
 Hollow cylinder of radii R_1, R_2 and mass m rotates about its axis	$\frac{1}{2} m (R_1^2 + R_2^2)$
 Solid disk of radius R and mass m rotates about its axis	$\frac{1}{2} m R^2$

Geometrical shape	Moment of inertia about the defined axis
 Rod of mass m rotates about its middle	$\frac{1}{12} m L^2$
 Rod of mass m rotates about its end	$\frac{1}{3} m L^2$
 Plate of mass m rotates about its middle	$\frac{1}{12} m (a^2 + b^2)$